

A Stochastic Extended Path to HMC pipeline

Mátyás Farkas

International Monetary Fund

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Views expressed are my own and do not necessarily represent the IMF.

Main Takeaway First

Use the linear solution unless the likelihood needs the nonlinear correction.

What I do

Each period starts from ROM1. A neural net adds only the SEP-ROM1 residual, and only where the training support says that correction is reliable.

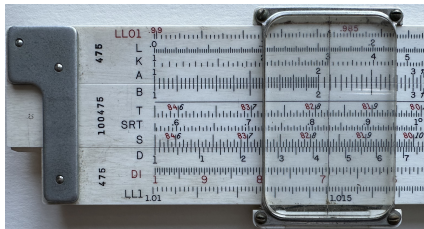
Why it matters

The nonlinear solver is not called inside HMC. The sampler sees one smooth likelihood, while the likelihood records whether each period used ROM1 or ROM1+NN.

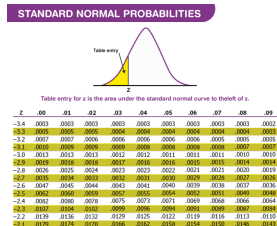
Frontier contribution: a differentiable likelihood whose solution order is observed period by period: linear where it is enough, nonlinear where it improves the likelihood, direct SEP as the teacher and benchmark.

The Intuition: Precompute the Hard Object

A surrogate is useful only when we are clear about the hard function it replaces.



Logs and products were once handled by tables and interpolation.



Hard functions were made operational by precomputation.

Here the hard function is the SEP transition map:

$$(s_t, \varepsilon_{t+1}, \theta) \mapsto s_{t+1}^{\text{SEP}}.$$

Images adapted from the previous Nonlinear DSGE presentation. Slide-rule source: Arnold Reinhold, Wikimedia Commons. Z-score table source: BYJU'S.

The Hook

The economics wants nonlinear dynamics; the posterior sampler wants speed.

Economists care about nonlinear states.

- Effective lower bounds.
- Occasionally binding constraints.
- Large shocks and crisis periods.
- Curvature in pricing, wages, investment, and utilization.

But estimation usually retreats to linear tools.

- Perturbation is fast.
- Kalman filtering is stable.
- The full nonlinear likelihood is too expensive.
- Particle filters can be fragile in high dimension.

The goal is not to replace linear methods everywhere. It is to know when linear is enough and when the nonlinear correction changes inference.

Question

Can we move the nonlinear solve outside the posterior loop without losing the likelihood?

The empirical problem

Medium-scale DSGE models are useful because they connect mechanisms, shocks, and observables. The same structure makes nonlinear estimation expensive.

What we want

- Structural parameters with posterior uncertainty.
- Likelihood comparisons across nonlinear mechanisms.
- Period-level diagnostics that say when linearization is enough.

What blocks us

- Repeated nonlinear solves.
- Numerical failures away from the mode.
- Slow, noisy, or non-differentiable likelihood evaluations.

Where This Fits

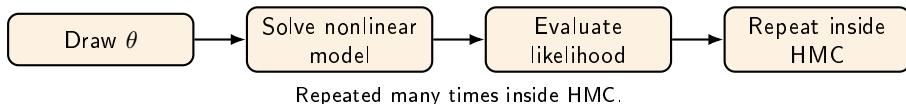
Approach	Main idea	Limitation for this talk
Particle-filter likelihoods Projection / OBC filtering	Integrate nonlinear latent states Approximate policy functions and split regimes	Accurate; costly and noisy in large systems Stable near constraints; still costly in estimation
Global NN solvers / MKR	Amortize nonlinear objects over states and parameters	Need likelihood validation and support checks
This paper	ROM1 by default, NN residual when supported	A smooth HMC likelihood with reported support and error

The neural net is not the model. It is the correction to the linear solution, trained against SEP and used only where checked.

Closest strands: particle-filter likelihoods; projection and piecewise OBC methods such as Gust et al.; and neural-network global solvers such as Kase, Melosi, and Rottner (MKR).

The Computational Problem

The nonlinear solve is too expensive to repeat at every HMC proposal.



The nested-loop bottleneck

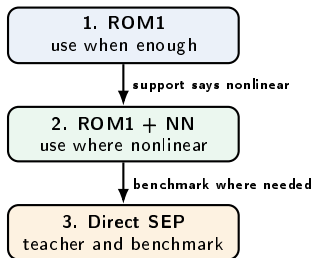
$$\theta \mapsto g_{\theta}^{\text{SEP}} \mapsto p(y_{1:T} \mid \theta) \mapsto \text{MCMC proposal}$$

The expensive object is recomputed at the point where the sampler needs thousands of fast, stable evaluations.

Benchmark approaches put nonlinear filtering or solution work inside the likelihood loop; see Fernández-Villaverde–Rubio-Ramírez (2007), Fernández-Villaverde–Rubio-Ramírez–Schorfheide (2016), and Herbst–Schorfheide (2016).

The Solution in One Picture

One model, one likelihood, three solution orders.



At each likelihood date

$$\hat{g}_t = g_{\theta}^{\text{ROM1}} + \omega(d_t) r_{\phi}(s_t, \varepsilon_{t+1}, \theta).$$

- d_t summarizes support, inversion, and regime diagnostics.
- $\omega(d_t) \approx 0$: ROM1 dominates.
- $\omega(d_t) \approx 1$: nonlinear residual is active.
- Outside validated support: flag the draw or benchmark with direct SEP.

The switch is inside the likelihood evaluation. The posterior draw does not choose a different model; it reveals where first-order dynamics suffice and where nonlinear solution information enters.

Algorithm: Switching Solution-Order Likelihood

The algorithm returns a likelihood value and a record of which solution order was used.

Stage	Operation	Comparison to related methods
Offline 1	Solve SEP at design triples (s, ε, θ) .	SEP is the expensive teacher, not the online likelihood engine.
Offline 2	Compute ROM1 at the same triples and train $r_\phi \approx g^{\text{SEP}} - g^{\text{ROM1}}$.	The NN learns only what the linear solution misses.
Online 1	Given θ , recover or sample shocks and compute support score d_t .	Filter-free shock treatment follows Childers et al.; inversion gives the medium-scale profile target.
Online 2	Evaluate $\hat{g}_t = g_\theta^{\text{ROM1}} + \omega(d_t)r_\phi$.	If $\omega_t \simeq 0$, this is linear; if $\omega_t \simeq 1$, the SEP residual is active.
Online 3	Accumulate likelihood and use HMC/NUTS on the differentiable target.	HMC sees one smooth target; the output records where nonlinear information enters.

Related methods: Fair–Taylor (1983) and Adjemian–Juillard (2013, 2025) for SEP; Childers et al. (2022) for filter-free HMC; Kase–Melosi–Rottner (2025) for neural-network surrogate estimation.

What Exactly Is Approximated?

The NN approximates a residual map, not preferences, constraints, or equilibrium conditions.

Equilibrium discipline

$$F(s_{t-1}, s_t, s_{t+1}, \varepsilon_t; \theta) = 0$$

is imposed by SEP offline, including expectations and constraints.

Training object

$$g_{\theta}^{\text{SEP}}(s_t, \varepsilon_{t+1}) = s_{t+1}^{\text{SEP}}.$$

Residual target

$$r_{\phi}(s, \varepsilon, \theta) \approx g_{\theta}^{\text{SEP}}(s, \varepsilon) - g_{\theta}^{\text{ROM1}}(s, \varepsilon).$$

Online map

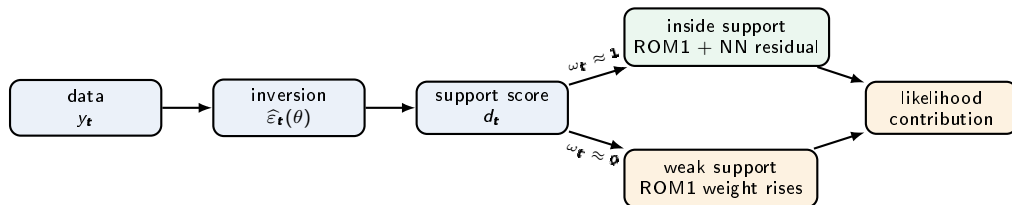
$$\hat{g}_{\theta, \phi} = g_{\theta}^{\text{ROM1}} + \omega(s, \varepsilon, \theta) r_{\phi}.$$

If ROM1 is exact, the residual is zero. If nonlinear dynamics matter, the residual is precisely the object the NN has to learn.

Relative to deep equilibrium nets and NN policy-function solvers (Maliar–Maliar–Winant, 2021; Azinovic–Gaegauf–Scheidegger, 2022), the network here emulates the SEP-implied transition residual used by the likelihood.

How the Switching Rule Operates

The gate adds the NN correction only on checked support.



Switching transition map

$$\hat{g}_t = g_{\theta}^{\text{ROM1}} + \omega_t r_{\phi}, \quad 0 \leq \omega_t \leq 1.$$

What the switch reports

- ROM1 periods.
- SEP-ROM1 correction periods.
- Weak-support periods to benchmark.

Output: likelihood contribution plus period-level solution-order diagnostic.

Step 0: Choose the Estimation Target

First define the likelihood; then decide how to approximate its transition map.

$$s_{t+1} = g_{\theta}(s_t, \varepsilon_{t+1}), \quad y_t = h(s_t; \theta) + \eta_t, \quad \varepsilon_t \sim \mathcal{N}(0, I), \quad \eta_t \sim \mathcal{N}(0, \Sigma_y).$$

Objects

- Structural parameters θ .
- State vector s_t .
- Structural shocks ε_t .
- Observables y_t .

Choice to make explicit

- Direct measurement-error likelihood or inversion/profile criterion.
- Which parameters vary in training and HMC.
- Parameter and shock support.
- Which regions are checked against direct SEP.

A surrogate that is accurate for one target need not be accurate for another. This is why the validation is tied to the likelihood, not only to one-step forecasts.

The target notation follows the nonlinear state-space estimation tradition in Fernández-Villaverde–Rubio–Ramírez (2007), Fernández-Villaverde–Rubio–Ramírez–Schorfheide (2016), and Herbst–Schorfheide (2016).

Step 1: Solve the Nonlinear Model Offline

Use SEP where accuracy matters most: to create the training target and benchmark cells.

For sampled triples (s, ε, θ) , compute the nonlinear transition

$$s_{\text{SEP}}^+ = g_{\theta}^{\text{SEP}}(s, \varepsilon).$$

$$\mathbb{E}_t m(s_{t+1}) \approx \sum_{j=1}^J w_j m(g_{\theta}^{\text{SEP}}(s_t, \varepsilon_{j,t+1})).$$

Why SEP?

- Keeps the nonlinear equations.
- Handles occasionally binding constraints.
- Produces the solved transition that ROM1 and the NN are compared to.

What must be reported

- Solver tolerances.
- Failed and successful points.
- Shock and parameter support.
- Initialization strategy.

SEP builds on Fair–Taylor (1983) and the stochastic extended path implementation of Adjemian–Juillard (2013, 2025). Here SEP is used as an offline teacher and benchmark, not as the repeated likelihood evaluator.

Step 2: Compute the Linear Baseline at the Same Points

Linearization is the control variate: cheap, stable, and often already right.

At the same inputs, compute

$$s_{\text{ROM1}}^+ = g_{\theta}^{\text{ROM1}}(s, \varepsilon).$$

Reason

ROM1 is fast and often accurate near the steady state. The NN should learn only the forecast error of ROM1 relative to SEP.

Nonlinear object = linear baseline + learned nonlinear correction.

ROM1 uses the perturbation/linear rational-expectations workhorse logic associated with Sims (2002) and the perturbation literature; here it is a control variate and fallback, not the final nonlinear estimator.

Step 3: Train the Residual Surrogate

Train the smallest object that carries the nonlinearity.

Train r_ϕ on the residual

$$r_\phi(s, \varepsilon, \theta) \approx g_\theta^{\text{SEP}}(s, \varepsilon) - g_\theta^{\text{ROM1}}(s, \varepsilon).$$

Why residual learning

- Smaller target.
- Zero target where ROM1 is already right.
- Directly measures what linearization misses.

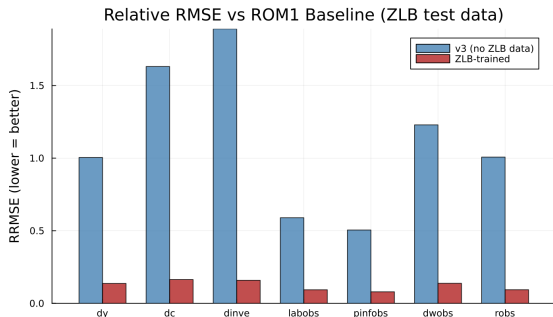
Resulting transition

$$g_\phi(s, \varepsilon, \theta) = g_\theta^{\text{ROM1}}(s, \varepsilon) + r_\phi(s, \varepsilon, \theta).$$

Compared with full-map neural solvers (Maliar–Maliar–Winant, 2021; Azinovic–Gaegauf–Scheidegger, 2022) and likelihood surrogates (Kase–Melosi–Rottner, 2025), the target is the smaller ROM1 residual.

Step 4: Validate the Surrogate Before Estimation

The gate can only be trusted where direct SEP checks say the residual is accurate.



Checks

- Held-out direct SEP points.
- RMSE and relative RMSE by observable.
- Regime-specific accuracy: binding, non-binding, near-boundary.
- Finite likelihood and finite-gradient checks for HMC.

Rule

Outside checked support, fall back toward ROM1 or benchmark the draw with direct SEP.

The validation logic follows surrogate-model practice: out-of-sample errors, support checks, and posterior-level diagnostics; compare Kennedy-O'Hagan (2001), Peherstorfer-Willcox-Gunzburger (2018), and HMC diagnostics in Hoffman-Gelman (2014).

Step 5: Build the Likelihood

The direct benchmark and the medium-scale target answer different questions.

Direct validation target

$$s_{t+1}^m = g_{\theta}^m(s_t^m, \varepsilon_{t+1}), \quad m \in \{\text{SEP}, \phi\}.$$

$$\ell_m(\theta, \varepsilon_{1:T}, s_0) = \sum_{t=1}^T \log \mathcal{N}(y_t; h(s_t^m; \theta), \Sigma_y).$$

This is the clean synthetic-data benchmark: does surrogate-HMC recover the same posterior as direct SEP-HMC?

The direct validation target uses the filter-free latent-shock formulation of Childers et al. (2022). The profiled target differs from particle/Kalman filtering by recovering shocks period by period and reporting the solution-order switch.

Medium-scale profiled target

$$\hat{\varepsilon}_{t+1}(\theta) = l_{\theta}(y_{t+1}, s_t),$$

$$\hat{s}_{t+1} = g_{\theta}^{\text{ROM1}}(s_t, \hat{\varepsilon}_{t+1}) + \omega_t r_{\phi}(s_t, \hat{\varepsilon}_{t+1}, \theta).$$

Here shocks are recovered period by period. The gate reports when the likelihood uses the nonlinear correction rather than ROM1.

Step 6: Sample Parameters with HMC

Filter-free HMC moves parameters and shocks together through a differentiable recursion.

$$q = (\theta, s_0, \varepsilon_{1:T}), \quad \log \pi_m(q \mid y_{1:T}) = \log p(\theta) + \log p(s_0 \mid \theta) + \sum_{t=1}^T \log p(\varepsilon_t) + \ell_m(q).$$

What is sampled

- Structural parameters θ .
- Initial condition or state objects.
- Latent structural shocks $\varepsilon_{1:T}$.
- Same prior and transforms in direct and surrogate runs.

What HMC tests

- The recursion is differentiable.
- Gradients are available through $\hat{g}_t$.
- Divergences, energy behavior, and mixing test local geometry.
- Matched settings make direct and surrogate posteriors comparable.

For validation, a low surrogate RMSE is not enough. The posterior from surrogate-HMC must agree with the posterior from direct SEP-HMC within Monte Carlo error.

Filter-free per-period likelihood evaluation follows Childers et al. (2022); NUTS/HMC diagnostics follow Hoffman–Gelman (2014) and Betancourt (2017).

Diagnostics: Report What the Switch Uses

The likelihood should leave an audit trail: where was it linear, where nonlinear, and where unsupported?

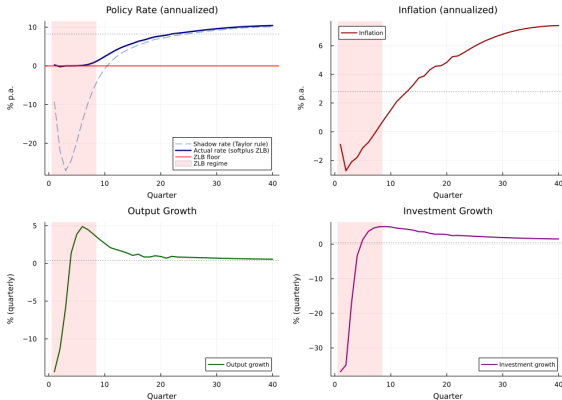
Object	Question	Failure mode
SEP support	Did the solver cover the region?	Hidden selection
Surrogate fit	Is the map accurate where used?	Biased likelihood
Gradients	Are target derivatives finite?	Bad HMC geometry
HMC chains	Do chains mix and agree?	Unreliable posterior
Direct SEP checks	Does SEP confirm key cells?	Unsupported extrapolation

The replication package should make these checks mechanical: every likelihood run should report support, gate weights, residual error, gradients, and HMC behavior.

These diagnostics distinguish the method from black-box neural estimation: the likelihood reports support, residual error, gradients, HMC behavior, and direct SEP checks where feasible.

Validation: A Hard Nonlinearity

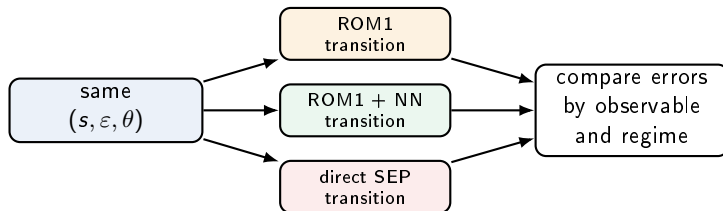
Validation is meaningful only where the nonlinear model visibly differs from ROM1.



Same shocks, nonlinear occasionally-binding model versus linearized model.

Validation: Same Inputs, Three Objects

Hold the inputs fixed, then ask what each solution order predicts.



Accuracy question

$$\|g^{\text{SEP}} - [g^{\text{ROM1}} + r_\phi]\| \ll \|g^{\text{SEP}} - g^{\text{ROM1}}\|.$$

Posterior question

$$p_\phi(\theta \mid y_{1:T}) \approx p_{\text{SEP}}(\theta \mid y_{1:T}).$$

The first question tests the map. The second tests the likelihood. A conference audience should see both.

Validation: What the Small Model Shows

The small model asks whether the full estimation loop can recover known nonlinear parameters.

Design

- Nonlinear New Keynesian prototype.
- Synthetic data generated by the nonlinear model.
- Selected shock-scale parameters estimated.
- Direct SEP-HMC compared to surrogate-HMC.

Interpretation

- The pipeline can recover parameters when the data are generated by the nonlinear model.
- The validation is local to the model and support.
- It is a feasibility proof, not a universal theorem.

The medium-scale laboratory follows the Smets–Wouters tradition and the nonlinear/ZLB estimation literature around Gust–Herbst–López-Salido–Smith (2017). Results on this slide are preliminary diagnostics.

Preliminary Application: Smets–Wouters

Smets–Wouters is the stress test: many states, many shocks, and several nonlinear mechanisms.

Empirical laboratory

Smets–Wouters style model with investment adjustment costs, variable utilization, Kimball pricing and wages, and occasionally binding constraints.

Why this is useful

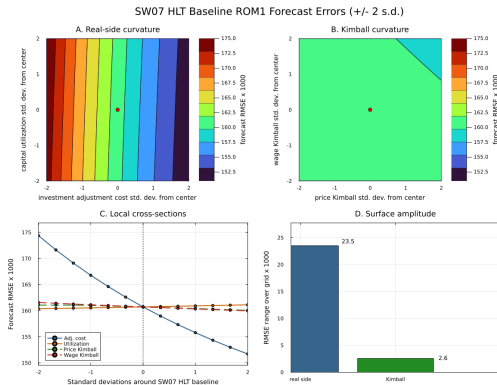
- Standard empirical DSGE laboratory.
- Rich enough to stress nonlinear mechanisms.
- Mechanisms are interpretable: investment, utilization, pricing, wages, constraints.

Why this is hard

- Higher-dimensional state.
- Fragile nonlinear solves.
- Strong need to know whether posterior draws stay inside checked support.

Preliminary HLT Evidence

Near the posterior region, ROM1's largest SEP error comes from real-side curvature.

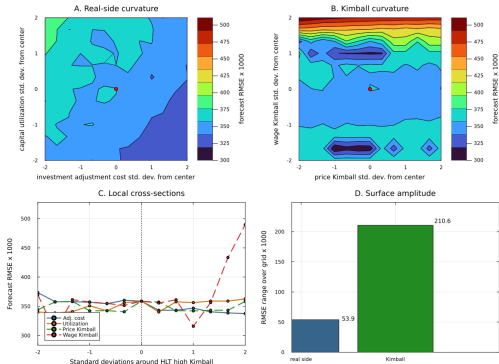


Preliminary diagnostic evidence: investment and utilization curvature move the SEP-ROM1 forecast error more than Kimball curvature around this posterior-region neighborhood.

Preliminary Stress Test: Kimball Can Matter Away From the Posterior Region

Kimball curvature matters in a high-curvature neighborhood, but that is not the posterior region.

HLT High Kimball ROM1 Forecast Errors (± 2 s.d.)



What changes

If the calibration is deliberately moved to a high price/wage-curvature neighborhood, Kimball curvature can dominate the local forecast-error surface.

What this means

The posterior-region result says where the likelihood needs nonlinear help in this estimation problem. It is not a claim that Kimball curvature never matters.

Computational Accounting

The computational gain comes from paying the nonlinear price only once, then using it selectively.

Task	Solution order	Role
SEP solves	benchmark order	expensive teacher
ROM1 solution	first order	cheap default
ROM1 + NN residual	switched-in order	nonlinear correction
Gate/support rule	reporting rule	chooses order by date
HMC	posterior sampler	explores switched target

The online estimator is cheap because ROM1 carries the easy part.
The NN is valuable because it adds only the nonlinear part SEP teaches offline.

Preliminary Evidence: What the Diagnostics Say

The current evidence supports the pipeline, while the full posterior run is still being extended.

Method diagnostics

- Corrected unit-shock HLT MVP: 3,200 direct SEP training points.
- ROM1-residual NN cuts observable residual RMSE by 87–94 percent.
- Small-model ELB/OBC matched HMC diagnostics.
- Reduced HLT bridge grid: direct SEP, ROM1, and surrogate surfaces.
- Full-NN and linear+gate MVP HMC smokes run without divergences.

Interpretation

- The switching likelihood is operational in controlled settings.
- The medium-scale evidence should be read as diagnostic, not final.
- The full 18-parameter production posterior is still being extended.
- The relevant question is whether the posterior lies on checked support.

Closing Takeaways

A good nonlinear DSGE estimator should know when not to be nonlinear.

1. The computational obstacle is clear: global nonlinear solves are too expensive for the inner likelihood loop.
2. The solution is simple: ROM1 does the cheap part; the NN adds the SEP residual only where support has been checked.
3. Direct SEP remains the teacher and benchmark, not the online estimator.
4. The small nonlinear validation checks posterior agreement, not only map fit.
5. The preliminary Smets–Wouters evidence is useful because it is tied to solver support, residual errors, gradients, and HMC diagnostics.

For this audience, the key object is not a black-box NN. It is a structural likelihood whose solution order is observed, gated, and benchmarked.

Thank you

Questions and comments welcome.

Backup: Relation to Gust et al. and MKR

Approach	What is approximated?	How inference is done
Gust et al. style OBC estimation	Policy functions with fixed projection basis and regime handling	Particle-filter likelihood and particle-filter MCMC
Kase–Melosi–Rottner	Neural-network objects over states and parameters	Fast estimation after offline amortization
This deck	SEP transition residual over ROM1	Differentiable switched likelihood for HMC

The shared idea is amortization. The distinction is the likelihood object: I train the SEP–ROM1 residual, keep ROM1 as the default online transition, and report posterior-support checks explicitly.

Backup: Runtime Budget

On this workstation, budget about one week for the production run.

Component	Measured anchor	Production budget
Base SEP data	1,664 corrected unit-shock samples in 2.8 hours	2–4 days for 125 parameter blocks at longer sample and horizon
ZLB SEP data	1,536 binding samples in 6.3 hours	2–5 days to obtain about 5,000 accepted binding points
Combine and train	3,200 points trained quickly; NN lowers ROM1 residual RMSE by 87–94 percent	Less than 2 hours unless the dataset is much larger
Surrogate HMC	100 warmup + 100 draws in 25 minutes at tree depth 5	3–8 hours per 1,500-iteration chain

If base and ZLB jobs run in parallel, the end-to-end wall-clock estimate is **5–8 days**, including training and matched HMC checks.

Backup: Target and Surrogate Error

$$e_{\phi}(s, \varepsilon, \theta) = g_{\theta}^{\text{SEP}}(s, \varepsilon) - [g_{\theta}^{\text{ROM1}}(s, \varepsilon) + r_{\phi}(s, \varepsilon, \theta)] .$$

Validation quantity

$$\text{RMSE}_{\text{val}} = \left(\frac{1}{N} \sum_{i=1}^N \|e_{\phi}(s_i, \varepsilon_i, \theta_i)\|^2 \right)^{1/2} .$$

The same held-out inputs are evaluated under direct SEP, ROM1, and the surrogate.

Backup: Matched HMC Validation Design

- Same synthetic data.
- Same prior and parameter transforms.
- Same initial values and random seeds.
- Same NUTS settings: warmup, draws, target acceptance, tree depth.
- Compare posterior means, MCSEs, and 90 percent intervals.

Pass criterion

The surrogate posterior should overlap the direct SEP posterior within Monte Carlo uncertainty, with no systematic directional distortion.

Backup: Why SEP?

- Keeps the original nonlinear equations.
- Handles occasionally binding constraints directly.
- Produces a simulation object that can be compared with perturbation.
- Expensive, but parallel over training points.

Tradeoff

SEP is a good offline teacher, but a poor online likelihood engine.

Backup: Posterior Interpretation

- The small-model validation targets direct likelihood comparisons.
- The medium-scale empirical exercise uses an inversion/profiled criterion.
- Surrogate HMC results should be read as criterion-based posterior diagnostics until a full direct-SEP HMC benchmark is feasible.

The method is promising because the validation pieces work, not because every medium-scale empirical caveat is already solved.